



A closed-form analytical model for estimating permeate salinity in reverse osmosis systems

Mingheng Li ^{a,*,}, Joseph Li ^b

^a Department of Chemical and Materials Engineering, California State Polytechnic University, Pomona, CA 91768, USA

^b Department of Chemical and Biomolecular Engineering, University of California, Berkeley, CA 94720, USA

ARTICLE INFO

Keywords:

Reverse osmosis
Permeate salinity
Analytical model
Desalination
Water flux
Concentration polarization

ABSTRACT

This study presents a simplified yet accurate analytical expression for estimating permeate salt concentration in reverse osmosis (RO) systems. The model is derived from classical solution-diffusion theory for a single membrane element and extended to multi-element systems using effective flux approximations. It is validated against five datasets: two seawater RO datasets based on single-element experiments conducted under varying recovery, feed concentration, and temperature (one recent and one historical); a brackish water module using cellulose acetate membranes; a full-scale brackish water RO plant treating real feedwater with complex ionic composition; and simulation data generated using ROSA (DuPont WAVE) for full-scale brackish and seawater systems. In all cases, the model shows strong agreement with reference data, with relative errors typically below 3 percent. Rather than replacing commercial design software, the proposed approach offers a fast, transparent, and straightforward method for estimating permeate concentration, suitable for use alongside commercial tools or when a quick, reasonably accurate estimate is needed. These parameters are generally available from design specifications or system monitoring, making the model especially useful for rapid design screening, real-time performance diagnostics, and operator training.

1. Introduction

Reverse osmosis (RO) has become a cornerstone technology for seawater and brackish water desalination due to its high separation efficiency, operational reliability, and scalability [1,2]. Among the key performance indicators in RO systems, the salt concentration of the permeate is especially important, as it directly impacts product water quality, regulatory compliance, and post-treatment requirements [3].

While detailed process models can predict permeate concentration with high fidelity [4], they are often computationally intensive and rely on numerous input parameters that may not be readily available during system design or routine operation. This complexity can limit their applicability, particularly in resource-constrained or time-sensitive settings.

In many practical scenarios such as preliminary design, system diagnostics, or control system tuning, engineers and operators require a quick and reasonably accurate method for estimating permeate salinity [5–8]. A simplified analytical expression can offer immediate insight into system behavior, supporting timely decision-making without the need for extensive simulation or experimentation [9,10].

Unlike proprietary simulation platforms provided by membrane manufacturers (e.g., DuPont WAVE and Toray DS), which operate as

black-box tools primarily intended for system design, the proposed analytical expression offers a transparent and physically grounded alternative. It requires only minimal input data, often obtainable from routine field measurements such as recovery and lead/tail flux, and is well suited for use in real-time diagnostics, operator training, and integration into automated control systems where full numerical simulation may not be practical. While permeate salinity is often monitored using conductivity sensors at the final product outlet, such measurements alone may not fully capture performance drift or membrane degradation. The proposed model can serve as a soft sensor by providing a baseline prediction of permeate concentration based on operating conditions. Significant deviations between measured and predicted values may indicate fouling, aging, or other anomalies. This diagnostic capability applies to both brackish and seawater RO systems and is particularly useful when distributed sensors or detailed modeling are unavailable.

Motivated by this need, the present work proposes a simple yet effective analytical equation for estimating permeate salt concentration in both seawater and brackish water RO systems. The model is derived from classical solution-diffusion theory for a single element and extended to multi-element configurations. It aims to strike

* Corresponding author.

E-mail address: minghengli@cpp.edu (M. Li).

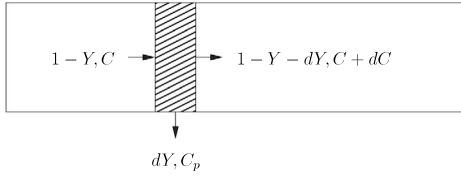


Fig. 1. Control volume for microscopic balance of water and salt.

a balance between simplicity and predictive accuracy, enabling practical application in field operations as well as broader system-level evaluations.

2. Mathematical model

2.1. Single element

According to classical solution-diffusion theory, the instantaneous salt concentration on the permeate side is:

$$\begin{aligned} C_p &= \frac{J_s}{J_w} \\ &= \frac{B_s \cdot \text{CPF} \cdot (C - C_p)}{J_w} \end{aligned} \quad (1)$$

where J_s and J_w are the salt and water fluxes, respectively, B_s is the salt permeability coefficient, C is the salt concentration in the bulk feed, and CPF is the concentration polarization factor. Rearranging yields:

$$C_p = \frac{B_s \cdot \text{CPF} \cdot C}{J_w + B_s \cdot \text{CPF}} \quad (2)$$

The corresponding instantaneous salt rejection (R) is:

$$\begin{aligned} R &= 1 - \frac{C_p}{C} \\ &= 1 - \frac{B_s \cdot \text{CPF}}{J_w + B_s \cdot \text{CPF}} \\ &\approx 1 - \frac{B_s \cdot \text{CPF}}{J_w} \end{aligned} \quad (3)$$

The approximation holds under typical RO operating conditions where $J_w \gg B_s \cdot \text{CPF}$, or $C \gg C_p$.

Consider a differential control volume within an RO module as illustrated in Fig. 1. At the inlet, the dimensionless flow rate and salt concentration are $1 - Y$ and C , respectively, where Y is the recovery ratio. At the outlet, these become $1 - Y - dY$ and $C + dC$. The permeate stream has flow dY and concentration C_p .

Applying a microscopic salt balance over the differential volume:

$$(1 - Y)C = (1 - Y - dY)(C + dC) + dY \cdot C_p \quad (4)$$

Simplifying yields:

$$(C - C_p)dY = (1 - Y)dC \quad (5)$$

Substituting R from Eq. (3):

$$R \cdot C \cdot dY = (1 - Y)dC \quad (6)$$

Assuming constant J_w , CPF, and B_s (reasonable for a single element), R is treated as constant. Integrating Eq. (6) using separation of variables:

$$R \int_0^Y \frac{1}{1 - Y} dY = \int_{C_0}^{C_1} \frac{1}{C} dC \quad (7)$$

where C_1 is the brine concentration. Solving gives:

$$C_1 = C_0(1 - Y)^{-R} \quad (8)$$

From the overall salt balance:

$$C_0 = (1 - Y)C_1 + YC_{po} \quad (9)$$

where C_{po} is the concentration of the permeate product at the outlet.

Substituting for C_1 yields:

$$C_{po} = \frac{1 - (1 - Y)^{1-R}}{Y} C_0 \quad (10)$$

This matches Equation (VIII-27) from Mulder's textbook [11].

Salt rejection R is not an intrinsic membrane property; it depends on both membrane properties and operating conditions. Substituting Eq. (3) into Eq. (10) gives:

$$C_{po} = \frac{1 - (1 - Y)^{\frac{B_s \cdot \text{CPF}}{J_w}}}{Y} C_0 \quad (11)$$

or in logarithmic form:

$$\frac{\ln(1 - C_{po}Y/C_0)}{\ln(1 - Y)} = \frac{B_s \cdot \text{CPF}}{J_w} \quad (12)$$

Thus, plotting $\ln(1 - C_{po}Y/C_0)/\ln(1 - Y)$ against CPF/J_w yields a straight line through the origin, with slope B_s . Since $C_{po}Y/C_0 \ll 1$, we can approximate $\ln(1 - C_{po}Y/C_0) \approx -C_{po}Y/C_0$, resulting in:

$$C_{po} = B_s \cdot C_0 \cdot \frac{[-\ln(1 - Y)/Y] \cdot \text{CPF}}{J_w} \quad (13)$$

The salt permeability coefficient B_s plays a central role in the model and can be estimated via calibration using measured permeate concentrations, or drawn from literature values for the specific membrane and solute system. While B_s reflects solute-specific and membrane-specific properties, the model remains robust as long as consistent calibration or reference values are applied for a given solute (e.g., NaCl) and membrane type under stable operating conditions.

This expression is consistent with the equation obtained in our earlier work using a different approach [12]. It can also be derived from Eq. (11) by applying the approximation $(1 - Y)^{\frac{B_s \cdot \text{CPF}}{J_w}} \approx 1 + \frac{B_s \cdot \text{CPF}}{J_w} \ln(1 - Y)$ which holds under the assumption that $\frac{B_s \cdot \text{CPF}}{J_w} \ll 1$.

FilmTec recommends the empirical relation $\text{CPF} = \exp(0.7Y)$ for estimating element-level concentration polarization in RO systems [13], where Y is the recovery of an individual membrane element. The exponent 0.7 is an empirically fitted constant derived to best match pilot- and full-scale system performance. This simplified form originates from the more fundamental expression $\exp(J_w/k_m)$, where k_m is the mass transfer coefficient, and reflects typical hydraulic conditions and flow geometry in standard spiral-wound elements (see our prior work [14] for a comparison of both approaches). By relating CPF directly to recovery, a readily available design or operating parameter, this empirical model offers a convenient and robust surrogate for detailed hydrodynamic modeling, enabling rapid CPF estimation during preliminary design and field evaluations.

2.2. Multiple elements in series

In commercial-scale RO systems, multiple membrane elements (or modules) are installed in series within a pressure vessel. For the i th element, the retentate concentration at its outlet, C_i , can be expressed as:

$$C_i = C_{i-1}(1 - Y_i)^{-R_i} \quad (14)$$

where Y_i and R_i are the recovery and rejection terms for the i th element. The retentate concentration at the vessel exit is:

$$C_N = C_0 \prod_{i=1}^N (1 - Y_i)^{-R_i} \quad (15)$$

where N is the total number of elements in the vessel. The overall stage recovery is:

$$Y = 1 - \prod_{i=1}^N (1 - Y_i) \quad (16)$$

A stage-level salt balance gives the average concentration of the permeate product:

$$\begin{aligned}
 C_{po} &= \frac{C_0 - (1 - Y)C_N}{Y} \\
 &= \frac{1 - (1 - Y) \prod_{i=1}^N (1 - Y_i)^{-R_i}}{Y} C_0 \\
 &= \frac{1 - \prod_{i=1}^N (1 - Y_i)^{1-R_i}}{Y} C_0 \\
 &= \frac{1 - \prod_{i=1}^N (1 - Y_i)^{\frac{B_s \cdot \text{CPF}_i}{J_{w_i}}}}{Y} C_0
 \end{aligned} \quad (17)$$

Regrouping terms and taking logarithms yield:

$$\ln(1 - C_{po}Y/C_0) = \sum_{i=1}^N \frac{B_s \cdot \text{CPF}_i}{J_{w_i}} \ln(1 - Y_i) \quad (18)$$

Assuming $\ln(1 - C_{po}Y/C_0) \approx -C_{po}Y/C_0$, the expression simplifies to:

$$C_{po} = \frac{B_s}{Y} C_0 \sum_{i=1}^N \left[\frac{-\ln(1 - Y_i) \cdot \text{CPF}_i}{J_{w_i}} \right] \quad (19)$$

This is a natural generalization of the single-element formula, accounting for spatial variation in flux and concentration polarization. In the ideal case where B_s , CPF_i , and J_{w_i} are constant (i.e., uniform rejection R_i), Eq. (19) reduces to the single-element result, since $\sum_{i=1}^N \ln(1 - Y_i) = \ln(1 - Y)$.

However, in real systems, J_w varies along the length of a pressure vessel due to pressure drop and the progressive increase in osmotic pressure. In brackish water RO (BWRO), the tail-to-lead flux ratio typically ranges from 70% to 80%, following a quasi-linear decline along the vessel. In contrast, seawater RO (SWRO) often exhibits a steeper, slightly curved flux decay, with the tail-to-lead ratio dropping to as low as 20%. While CPF may increase slightly in BWRO, it generally decreases in SWRO due to the pronounced flux decline. Overall, variation in CPF is far less significant than that of flux. As a result, flux variation dominates the nonlinearity in Eq. (19).

In practice, Eq. (19) can be approximated using limited but commonly available plant data. The overall recovery Y and average flux $\bar{J}_w$ are typically known from system design or operation. The lead and tail fluxes (J_{w_1} , J_{w_N}), and sometimes their corresponding recoveries, can often be inferred from plant pressure and flow readings. Furthermore, commercial membrane elements are typically operated at element-level recoveries below 15%, which helps ensure that simplifying assumptions remain valid.

To simplify Eq. (19), we approximate the summation by evaluating the average behavior of each component term:

For $-\ln(1 - Y_i)$:

$$\begin{aligned}
 -\ln(1 - Y_i) &= \frac{1}{N} \sum_{i=1}^N -\ln(1 - Y_i) \\
 &= -\frac{1}{N} \ln(1 - Y) \\
 &= -\ln(1 - \bar{Y})
 \end{aligned} \quad (20)$$

where $\bar{Y}$ is the ideal constant per-element recovery required to maintain overall recovery Y :

$$\bar{Y} = 1 - (1 - Y)^{1/N} \quad (21)$$

Note that $1 - \bar{Y}$ is the geometric mean of $1 - Y_i$ since $(1 - \bar{Y})^N = 1 - Y = \prod_{i=1}^N (1 - Y_i)$.

For CPF_i , it is assumed that it follows the same expression as the single-element expression, given the fact that the variation in CPF_i is mild:

$$\text{CPF} = \exp(0.7\bar{Y}) \quad (22)$$

For $1/J_{w_i}$, the spatial profile of J_w is taken into account. It is observed that under typical design and operation conditions, there is approximately linear decay from the lead to tail element in a vessel consisting of 7 elements in BWRO: $J_w(x) = J_w(0)(1 - kx/L)$, where x is distance from inlet, L is total length of vessel, and $k = 1 - J_w(L)/J_w(0)$ is a constant parameter. The spatial mean of $1/J_w$ is:

$$\begin{aligned}
 \frac{1}{J_w} &= \frac{1}{L} \int_0^L \frac{1}{J_w(0)(1 - kx/L)} dx \\
 &= \frac{\ln\left(\frac{1}{1 - k}\right)}{J_w(0)k} \\
 &= \frac{\ln(J_w(0)/J_w(L))}{J_w(0) - J_w(L)} \\
 &= \frac{1}{J_{w,\text{lm}}}
 \end{aligned} \quad (23)$$

where $J_{w,\text{lm}}$ is the log-mean flux.

Accordingly, the flux term in Eq. (13) can be replaced by the log-mean flux $J_{w,\text{lm}}$ to account for spatial variation across the vessel. When variation is small, $J_{w,\text{lm}} \approx \bar{J}_w$, and the correction becomes negligible.

The effective, log-mean flux $J_{w,\text{lm}}$ can be calculated based on lead and rear-end fluxes:

$$J_{w,\text{lm}} = \frac{J_{w_1} - J_{w_N}}{\ln\left(\frac{J_{w_1}}{J_{w_N}}\right)} \quad (24)$$

which is not exactly the inverse of the mean flux $\bar{J}_w$:

$$\bar{J}_w = \frac{1}{N} \sum_{i=1}^N J_{w_i} \quad (25)$$

Based on the above, Eq. (19) can be estimated by:

$$\begin{aligned}
 C_{po} &= B_s \cdot C_0 \cdot \frac{[-\ln(1 - Y)/Y] \cdot \text{CPF}}{J_{w,\text{lm}}} \\
 &= B_s \cdot C_0 \cdot \frac{[-\ln(1 - Y)/Y] \cdot \text{CPF}}{\bar{J}_w} \cdot \frac{\bar{J}_w}{J_{w,\text{lm}}}
 \end{aligned} \quad (26)$$

It is important to note that, in general, the sum of products is not equal to the product of the averages. Accordingly, Eqs. (19) and (26) are not mathematically equivalent in a strict analytical sense. However, in the specific context of RO system design, Eq. (26) offers a highly accurate approximation due to practical constraints. Membrane manufacturers typically limit individual element recovery to below 15%–20%, and the overall recovery in a pressure vessel containing 6–7 elements rarely exceeds 50%–60%. Under these design guidelines, the variation in CPF and flux is smooth enough that the log-mean approximation performs well.

Despite being derived for BWRO systems, Eq. (26) also performs well in SWRO applications, where the flux profile is more strongly curved. Although the lead-to-tail flux ratio is larger, the recovery of tail-end elements is correspondingly lower. A more detailed justification for the log-mean approximation is provided in Appendix A. In short, the log-mean flux arises as a weighted harmonic mean of local fluxes when extending the single-element model to a multi-element configuration. When J_w decays linearly, the harmonic mean and log-mean are mathematically identical. More generally, when J_w decays quadratically (either convex or concave), the approximation still holds if the flux is a quasi-linear function of the cumulative sum of weights. Thus, Eq. (26) remains valid for a wide range of smooth, realistic flux profiles encountered in RO systems.

This approximation builds directly on the single-element expression (Eq. (13)), with a flux correction introduced via the log-mean to represent spatial variation along the vessel. In the special case of uniform flux, Eqs. (26) and (13) yield essentially the same result. The correction factor $\bar{J}_w/J_{w,\text{lm}}$ remains close to unity in BWRO systems and deviates more significantly in SWRO, reflecting the stronger flux decline. This

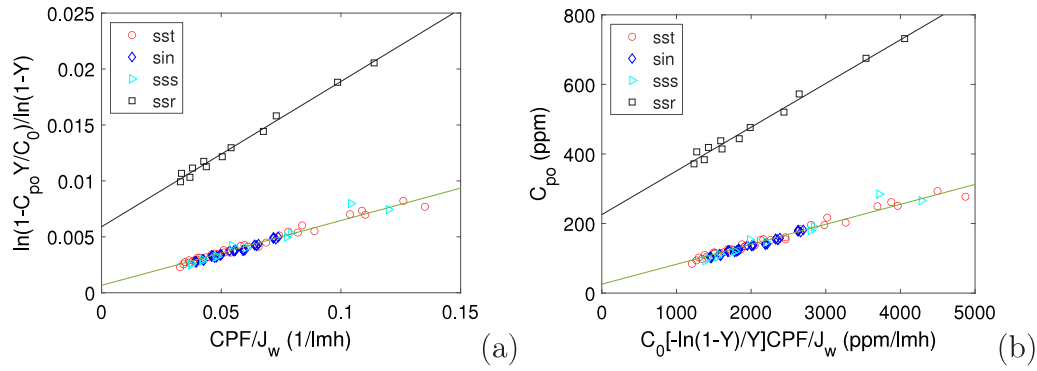


Fig. 2. Correlation between (a) $\ln(1 - C_{po} Y / C_0) / \ln(1 - Y)$ and CPF / J_w ($R^2 = 0.98-0.99$), and (b) C_{po} and $C_0 [-\ln(1 - Y) / Y] \cdot CPF / J_w$ [12]. Experimental data are from a single seawater spiral-wound element [15], with CPF estimated using $\exp(0.7Y)$.

approach strikes a balance between simplicity and predictive accuracy. Importantly, the estimation error remains bounded even under substantial flux variation. As demonstrated later, this approximation agrees with the full model in Eq. (19) within a 3% margin for representative BWRO and SWRO operating conditions.

To apply the proposed expression for estimating permeate salinity, the following input parameters are required: (i) feed concentration C_0 , (ii) recovery ratio Y , (iii) design-average water flux $\bar{J}_w$ and representative lead/tail element fluxes, (iv) an estimate of the concentration polarization factor (CPF), and (v) the salt permeability coefficient B_s . In most applications, C_0 , Y , and $\bar{J}_w$ are readily measurable or known from system design specifications or online monitoring systems. CPF can be approximated using empirical expressions such as $\exp(0.7Y)$, or refined based on hydrodynamic modeling. The parameter B_s may be estimated from literature data, manufacturer specifications, or calibration using measured permeate concentrations under known conditions. These parameters are typically accessible from design specifications or online monitoring systems, enabling use in diagnostics, monitoring, and rapid screening. The simplicity of required inputs enables implementation in design spreadsheets, digital twins, and soft sensor platforms for real-time performance tracking.

3. Results and discussion

In this section, single-element and full-scale data are used to validate equations presented in the previous section. All data are included in the supplemental material.

3.1. Seawater RO module under steady-state and sinusoidal feed conditions [15]

Experiments were conducted using a single FilmTec™ SW30-2540 element under both steady-state and sinusoidal feed conditions. The operating conditions included a temperature of approximately 27 °C, feed pressures ranging from 34 to 60 bar, a feed concentration of approximately 35,000 ppm, feed flow rates between 6 and 16 L/min, and recovery ratios from 3% to 23%. The CPF was estimated using the empirical relation $CPF = \exp(0.7Y)$ [13].

Fig. 2 compares the predictions of Eqs. (12) and (13) for four different test categories:

- **sst**: steady-state test,
- **sin**: sinusoidal feed test,
- **sss**: steady-state integrity check after sinusoidal testing,
- **ssr**: steady-state integrity check after rectified sinusoidal testing (indicative of membrane degradation).

Both plots show consistent trends, confirming the equivalence of Eqs. (12) and (13). The datasets are categorized into non-degraded (sst, sin, sss) and degraded (ssr) membranes. For non-degraded membranes, the model trained on sst data also fits sin and sss datasets well, with root mean square error (RMSE) values of 8.3, 6.7, and 15.5 ppm, respectively. For the degraded membrane condition (ssr), a separate correlation is necessary, with an RMSE of 12.3 ppm. Fig. 2 shows that the model trained on non-degraded membrane data (e.g., sst) fails to fit the degraded condition (ssr) without recalibration. This sensitivity enables the model to serve as a soft sensor for detecting deviations from baseline performance, providing early indication of fouling or degradation.

As noted in our prior work [12], the intercept in Fig. 2 is close to zero for datasets collected before membrane degradation but significantly elevated for those obtained after degradation. Eq. (13) does not include a constant offset and therefore predicts $C_p \rightarrow 0$ as $J_w \rightarrow \infty$, a known artifact of the simplified solution-diffusion framework. This behavior arises from the model's assumption that rejection approaches unity at very high flux. A more rigorous correction, such as Equation (78) in [16], introduces pressure-dependent solute transport and yields a finite asymptotic salt passage, consistent with observed membrane selectivity. Alternatively, the solution-friction framework [17,18] provides a physically motivated explanation by accounting for convective coupling between water and solute fluxes. Both approaches offer possible extensions to improve high-flux predictions beyond the current model's scope.

3.2. Seawater RO module at different temperatures and feed concentrations [19]

This study utilized a membrane assembly consisting of two feed channels and one permeate channel, sandwiched between two FilmTec™ FT30 SW membranes, each with an active area of 0.87 m × 1.17 m. Experiments were conducted under varying operating conditions, with temperatures ranging from 20 to 35 °C, feed pressures between 50 and 80 bar, feed concentrations from 25,000 to 40,000 ppm, flow rates of 7.5–13 L/min, and recovery ratios between 6.7% and 17.4%.

Fig. 3(a) shows the correlation results obtained using Eq. (13) under the assumption $CPF = \exp(0.7Y)$. Strong linear relationships were observed at all tested temperatures. The RMSE values were 9.6, 3.4, 5.2, and 9.7 ppm for 20, 25, 30, and 35 °C, respectively. The corresponding B_s values were 0.12, 0.15, 0.20, and 0.26 lmh, differing by less than 6% from those obtained via nonlinear regression using a Sherwood-type mass transfer correlation [20], where both the power-law indices and pre-exponential coefficients were optimized.

In addition, the relationship between B_s and temperature followed an Arrhenius-type expression: $\ln(B_s / [\text{lmh}]) = -4729.4 / (T / [^\circ\text{K}]) + 13.98$, as shown in Fig. 3(b). This result aligns with the well-documented trend

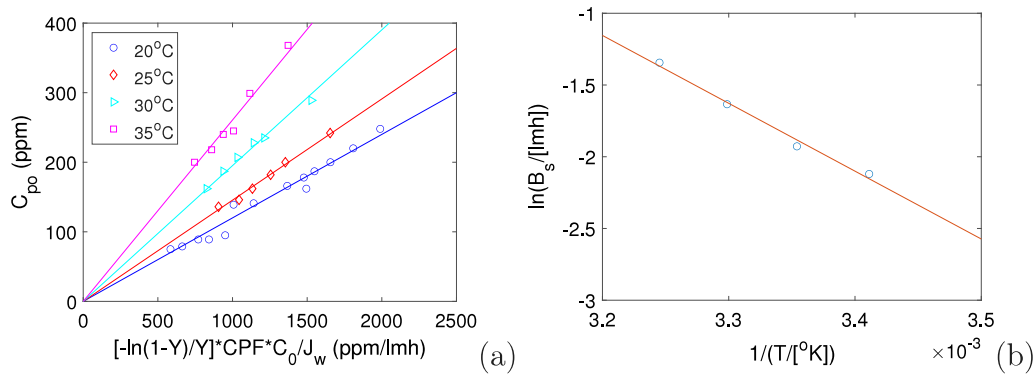


Fig. 3. (a) Correlation between C_{po} and $[-\ln(1-Y)/Y] \cdot CPF \cdot C_0 / J_w$ at different temperatures ($R^2 = 0.93$ – 0.96). (b) Arrhenius-type correlation between B_s and temperature ($R^2 = 0.99$). Experimental data are from a single seawater spiral-wound module [19].

that salt permeability increases with temperature. For example, under otherwise constant operating conditions, salt transport approximately doubles as the temperature increases from 20 to 35 °C. The Arrhenius-type dependence of B_s reported here is established specifically for NaCl. Extrapolation to other solutes (e.g., Ca^{2+} or SO_4^{2-}) requires separate validation.

Furthermore, when CPF values derived from nonlinear regression using Sherwood-type correlation [20] were plotted against Y on a semilogarithmic scale (Fig. 4), a clear linear relationship was observed. The slope values were 1.12, 1.02, 0.89, and 0.78 for 20, 25, 30, and 35 °C, respectively. The observed trend of decreasing CPF with increasing temperature is consistent with previous findings in the literature [21]. As discussed in our previous work [20], higher temperatures reduce the kinematic viscosity of the feed solution, which in turn increases the mass transfer coefficient and lowers the CPF under otherwise identical operating conditions.

The parameter B_s is typically determined by fitting the model to experimental data under an assumed CPF form. When the exponent in the empirical CPF expression is changed (e.g., from 0.7 to 1.02), the predicted C_{po} values remain nearly unchanged, but the fitted B_s shifts proportionally. This occurs because B_s and CPF always appear as a product in the correlation (e.g., Eq. (13)), and thus are not independently identifiable without separate measurement of mass transfer coefficients. In this sense, adjusting the CPF exponent effectively “rescales” the interpreted B_s , while preserving the accuracy of the model equation.

This sensitivity analysis does not indicate model instability, but rather underscores the coupling between fitted parameters and assumed CPF behavior. Regardless of the CPF exponent, the model maintains high predictive performance across the temperature range, as shown by consistent RMSE values and linearity in Fig. 3(a). Therefore, the correlation remains valid and useful for performance estimation across temperatures, as long as the CPF formulation is consistently applied. It is important to either (i) calibrate the CPF exponent for the specific membrane or (ii) treat it as a tunable parameter during model fitting, especially when the dataset spans a wide recovery range. Otherwise, extrapolation accuracy, particularly at high recoveries, may be compromised.

3.3. Brackish water RO module [22]

Experiments were conducted using a ROGA-4160HR spiral-wound module with cellulose acetate membranes developed by UOP, a design that is no longer commercially available. The operating conditions included a temperature of 25 °C, feed pressures ranging from 27 to 36 atm, feed concentrations slightly below 2000 ppm, flow rates between 5.4 and 31.7 L/min, and recovery ratios from 6.8% to 43.9%. It is worth noting that the recovery ratios used in some experiments reached up to 43.9% within a single module, which is significantly higher

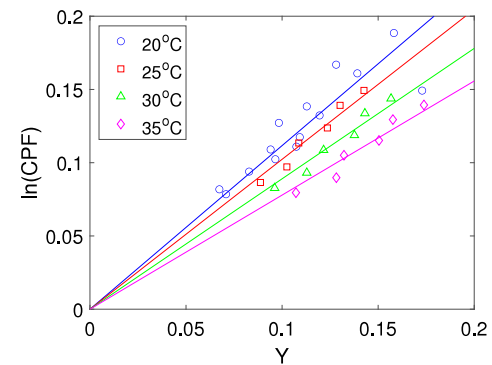


Fig. 4. Relationship between $\ln(CPF)$ and Y based on regression of experimental data [19], assuming a Sherwood-type correlation for the mass transfer coefficient [20].

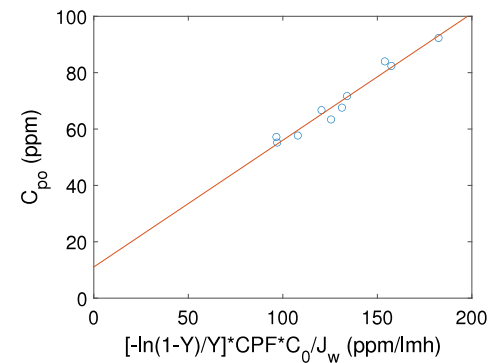


Fig. 5. Correlation between C_{po} and $[-\ln(1-Y)/Y] \cdot CPF \cdot C_0 / J_w$ ($R^2 = 0.96$). Experimental data are from a single cellulose acetate spiral-wound module [22].

than current design recommendations. Modern RO practice typically constrains element-level recovery to a low level (e.g., 15%) to reduce the risk of aggravated concentration polarization and fouling.

Assuming the empirical relation $CPF = \exp(0.7Y)$, Eq. (13) provided excellent agreement with the experimental data, as shown in Fig. 5, achieving an R^2 value of 0.96 and an RMSE of 2.3 ppm. The fitted line exhibited a slight positive intercept. A sensitivity analysis that varied the CPF exponent from 0.3 to 1.2 maintained R^2 values above 0.93, indicating the model’s robustness to variations in concentration polarization assumptions.

Although the ROGA module represents an earlier generation of membrane technology, the results support the applicability of the proposed correlation across different membrane chemistries and a broad range of operating conditions.

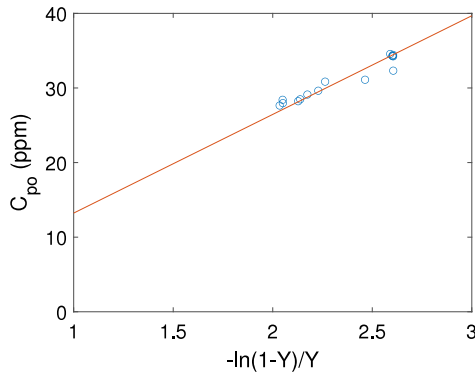


Fig. 6. Correlation between C_{po} and $-\ln(1-Y)/Y$ under constant design flux in a two-stage brackish water RO system ($R^2 = 0.93$).

3.4. Two-stage brackish water RO plant at constant design flux [23]

Field experiments were conducted at a full-scale brackish water desalination plant, as detailed in our prior work [23]. The feedwater used in this study was actual groundwater with a complex ionic composition, including multivalent ions, rather than a synthetic or simplified solution.

The system consisted of 294 FilmTec™ BW30-400 elements arranged in a two-stage (28:14) configuration, with seven elements per pressure vessel. It operated at a fixed permeate flow of 1235 gpm (280.5 m³/h), corresponding to an average system flux of 15.1 gfd (25.7 lmh), while the overall recovery ratio varied from 80% to 91%.

Under these constant-flux, variable-recovery conditions, a strong linear relationship was observed between C_{po} and $-\ln(1-Y)/Y$, as shown in Fig. 6. The proposed analytical correlation demonstrated excellent predictive accuracy, with an R^2 value of 0.94 and an RMSE of just 0.7 ppm. These results validate the robustness of the model under real-world conditions, including non-ideal water quality and large-scale system operation.

3.5. ROSA simulation data

Simulation data from two representative RO processes modeled using ROSA (now DuPont WAVE) are used to validate the proposed correlation at both the element and stage levels. The first scenario involves a two-stage BWRO system configured in a 2:1 array with seven BW30-400 elements per vessel, operated at 80% recovery and a design flux of 15.1 gfd (25.7 lmh) [23]. The second scenario consists of a one-stage SWRO system using six SW30HR-380 elements at 50% recovery and a flux of 8.4 gfd (14.3 lmh) [24]. The simulation results are summarized in Tables 1 and 2, where subscripts f , p , and i refer to feed, permeate, and element index, respectively.

Fig. 7 demonstrates that plotting C_{p_i} against $[-\ln(1-Y_i)/Y_i] \cdot \text{CPF}_i \cdot C_{f_i}/J_{w_i}$ in each element results in a straight line through the origin, validating Eq. (13). The slope corresponds to the salt permeability coefficient B_s , calculated as 0.2676 lmh for the full BWRO system (or 0.2739 and 0.2669 lmh for stages 1 and 2, respectively), and 0.0634 lmh for the SWRO system. In operational contexts, B_s can be treated as a calibrated parameter, provided the CPF approximation is applied consistently, as shown in sensitivity analyses in Section 3.2.

Fig. 8 shows that while Y_i varies significantly across elements, CPF_i remains relatively consistent, supporting the use of an average CPF in system-level approximations.

The element-wise contributions to stage-level permeate concentration are illustrated in Fig. 9. Differences in curve shapes between BWRO and SWRO reflect their distinct flux and recovery profiles. The average trend (red symbols) is computed using geometric averaging of $1-Y_i$ and log-mean flux, in accordance with Eqs. (21) and (24).

Table 1

ROSA simulation data for a two-stage BWRO at 80% recovery.

Element	Y_i	Q_{p_i} (gpm)	C_{p_i} (mg/l)	Q_{f_i} (gpm)	C_{f_i} (mg/l)	P_{f_i} (psig)
Stage 1:						
1	0.09	5.03	12.71	55.06	1257.56	225.99
2	0.1	4.85	14.43	50.02	1385.07	219
3	0.1	4.69	16.55	45.17	1532.52	212.93
4	0.11	4.55	19.17	40.47	1708.52	207.7
5	0.12	4.43	22.5	35.92	1922.85	203.25
6	0.14	4.32	26.84	31.49	2190.25	199.52
7	0.15	4.21	32.71	27.18	2533.92	196.46
Stage 2:						
8	0.09	4.01	37.17	45.93	2992.22	189.01
9	0.09	3.82	42.79	41.92	3274.84	183.58
10	0.10	3.63	49.56	38.11	3598.58	178.82
11	0.10	3.43	57.82	34.48	3971.55	174.68
12	0.10	3.23	67.98	31.05	4403.31	171.09
13	0.11	3.01	80.69	27.83	4905.38	168.01
14	0.11	2.79	96.79	24.81	5490.51	165.38

Table 2

ROSA simulation data for a one-stage SWRO at 50% recovery.

Element	Y_i	Q_{p_i} (gpm)	C_{p_i} (mg/l)	Q_{f_i} (gpm)	C_{f_i} (mg/l)	P_{f_i} (psig)
1	0.15	3.94	114.53	26.67	35 235.30	855.45
2	0.14	3.17	162.05	22.72	41 331.94	852.98
3	0.12	2.41	236.94	19.55	47 999.65	850.99
4	0.10	1.75	355.99	17.14	54 721.21	849.36
5	0.08	1.21	544.2	15.4	60 888.97	847.98
6	0.06	0.85	814.26	14.18	66 055.90	846.76

Table 3

Estimation of permeate product concentration from process data.

	BWRO (stage 1)	BWRO (stage 2)	SWRO
Y	0.58	0.52	0.50
N	7	7	6
$\bar{J}_w$ (lmh) (ROSA)	28.0	20.9	14.3
$\bar{Y}$ (Eq. (21))	0.117	0.100	0.109
CPF (Eq. (22))	1.09	1.07	1.08
$J_{w,lm}$ (lmh) (Eq. (24))	28.2	20.6	13.0
B_s (lmh)	0.2739	0.2669	0.0634
C_0 (mg/l) (ROSA)	1257.56	2992.22	35 235.30
C_{po} (mg/l) (ROSA)	20.3	59.5	263.3
C_{po} (mg/l) (Eq. (19))	20.4	59.5	256.2
C_{po} (mg/l) (Eq. (26))	19.9	58.9	257.4

Table 3 compares stage-level permeate concentration estimated using both the detailed element-wise model (Eq. (19)) and the simplified log-mean flux model (Eq. (26)) against the ROSA-calculated reference. In all cases, deviations remain within 3%.

Additional simulations spanning recoveries from 72%–90% (BWRO) and 35%–55% (SWRO) confirm that the simplified model remains accurate across a wide operating range. Fig. 10 shows that under constant design flux, both the ROSA simulations and the proposed model predict an approximately linear relationship between C_{po} and $-\ln(1-Y)/Y$, consistent with the trend observed in full-scale plant data (Fig. 6).

The analytical formulation developed in this work offers more than computational simplicity; it also provides physical insight into the mechanisms governing permeate salinity in RO systems. Specifically, the expression $C_{po} \propto B_s \cdot \frac{\text{CPF}}{J_w} \cdot \frac{\ln(1-Y)}{Y}$ shows that permeate concentration is determined by salt permeability (B_s), concentration polarization (CPF), water flux (J_w), and recovery ratio (Y). This explicit relationship reveals the competing influences of transport resistance and system recovery, which are often obscured in simulation outputs from platforms such as ROSA/WAVE.

By incorporating log-mean water flux, the model accounts for spatial variability across membrane elements and connects membrane-scale transport with stage-level system behavior. This bridging of scales enables not only fast estimation but also meaningful interpretation of

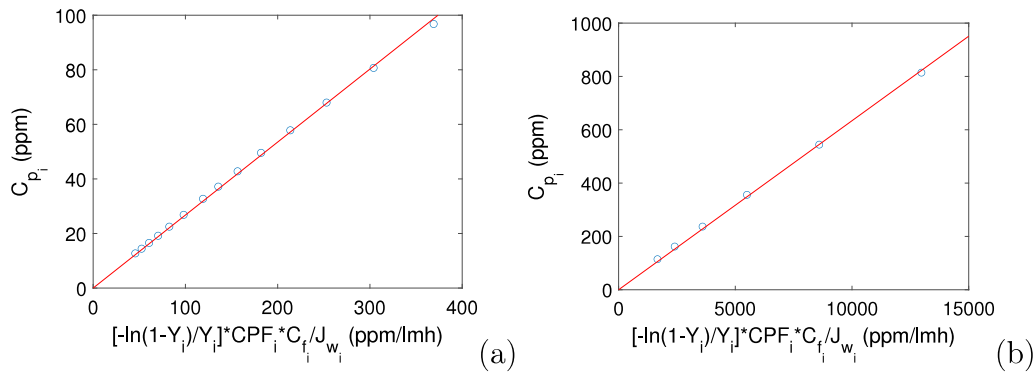


Fig. 7. Correlation between element-level C_{p_i} and $[-\ln(1-Y_i)/Y_i] \cdot CPF_i \cdot C_{f_i} / J_{w_i}$ in (a) two-stage BWRO and (b) one-stage SWRO using ROSA simulation data ($R^2 = 1.00$). CPF_i is calculated by $\exp(0.7Y_i)$.

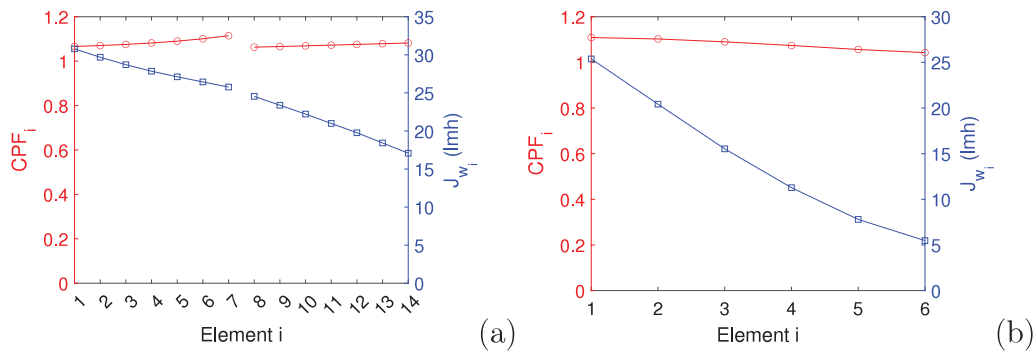


Fig. 8. Element-level CPF_i and J_{w_i} from ROSA for (a) BWRO (element 1–7 for stage 1 and 8–14 for stage 2) and (b) SWRO.

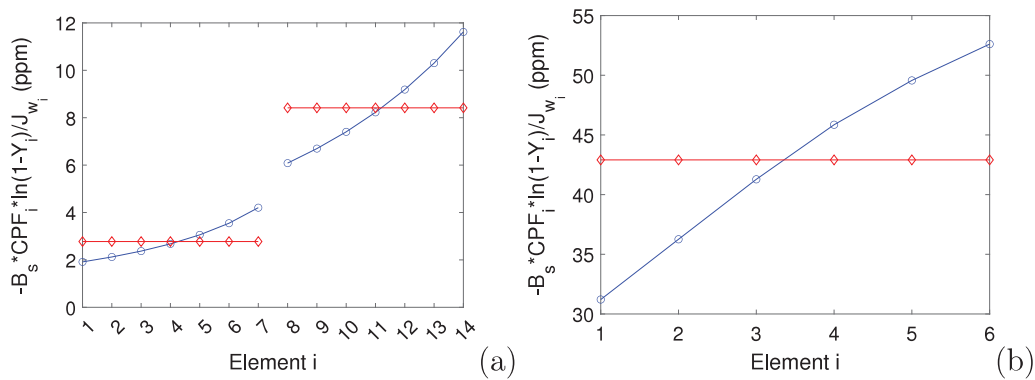


Fig. 9. Contribution of each RO element to the stage-level permeate product concentration (blue symbols) and the average representation (red symbols) for (a) BWRO (elements 1–7 in stage 1 and 8–14 in stage 2) and (b) SWRO.

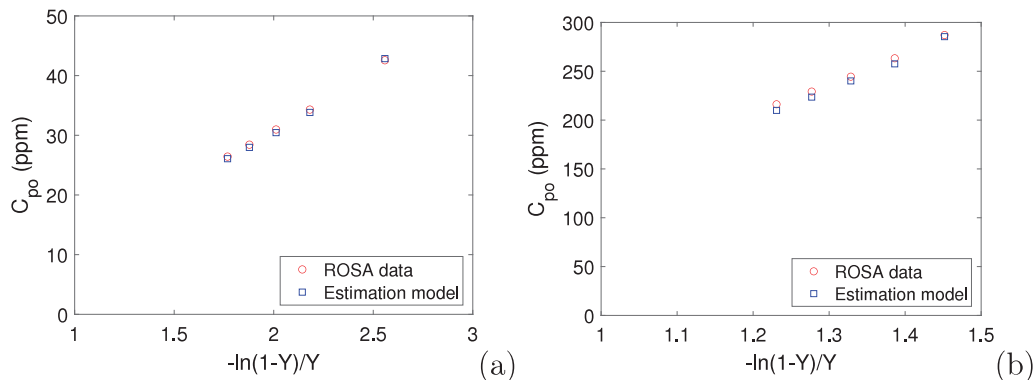


Fig. 10. Comparison between ROSA simulation results and short-cut model estimates for final permeate concentration (C_{p_o}) at varying recoveries under constant system-level design flux conditions for (a) a two-stage BWRO system and (b) a one-stage SWRO system.

performance trends, which can aid in diagnostics, optimization, and control of RO operations.

4. Conclusions

This work presents a physically transparent, closed-form model for estimating average permeate salt concentration in RO systems, requiring only several readily available inputs. Unlike simulation tools, the model offers interpretability, fast evaluation, and serves both design and operational diagnostic purposes.

The formulation, originally developed for a single membrane element, was extended to multiple elements in series within a pressure vessel. This extension captures spatial variations in water flux and concentration polarization by introducing an effective log-mean flux and a concentration polarization factor based on the geometric mean recovery. The resulting expression achieves a practical balance between physical accuracy and analytical simplicity.

The model was validated against five independent datasets, spanning both experimental and simulated scenarios: (1) a seawater RO element under variable recovery, (2) a seawater RO module tested across a range of temperatures and feed concentrations, (3) a historical brackish water RO module using cellulose acetate membranes, (4) a full-scale brackish water RO plant with variable recovery and constant design flux, and (5) element-wise data from ROSA/WAVE simulations for both two-stage BWRO and one-stage SWRO systems.

In all cases, the model demonstrated excellent predictive performance, with correlation coefficients (R^2) typically above 0.93. The analysis also confirmed that accurate predictions are achievable using the log-mean flux approach, even when only the fluxes at the lead and tail elements are available.

In addition to predictive accuracy, the model provides physical insights into system behavior, enabling users to better understand the influence of design and operating parameters on permeate salinity. It may also function as a soft sensor for membrane health monitoring by comparing predicted and measured salt concentrations, enabling early detection of fouling or degradation.

By relying only on parameters that are readily measurable or estimable in the field, the proposed framework enables quick and accurate estimation of permeate salinity. It is especially valuable for design screening, real-time monitoring, and operational diagnostics in both seawater and brackish water RO applications. Because the model avoids reliance on proprietary software and uses only basic inputs, it can be embedded in control platforms or used for rapid feasibility screening in early-stage RO system design.

CRedit authorship contribution statement

Mingheng Li: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Joseph Li:** Writing – original draft, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to acknowledge financial support from the National Science Foundation, United States (CBET-2140946). Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation.

Appendix A. Detailed justification for using log-mean flux in Eq. (26)

For N elements in a single pressure vessel:

$$\begin{aligned} C_{po} &= \frac{B_s}{Y} \cdot C_0 \cdot \sum_{i=1}^N \left[\frac{-\ln(1-Y_i) \cdot \text{CPF}_i}{J_{w_i}} \right] \\ &= \frac{B_s}{Y} \cdot C_0 \cdot \frac{-\ln(1-Y)}{\text{CPF}} \cdot \sum_{i=1}^N \left[\frac{-\ln(1-Y_i) \cdot \text{CPF}_i}{-\ln(1-Y_i) \cdot \frac{\text{CPF}}{\text{CPF}_i}} \cdot \frac{1}{J_{w_i}} \right] \\ &= \frac{B_s}{Y} \cdot C_0 \cdot \frac{-\ln(1-Y)}{N} \cdot \text{CPF} \cdot \sum_{i=1}^N \left[\frac{-\ln(1-Y_i) \cdot \text{CPF}_i}{-\ln(1-Y_i) \cdot \text{CPF}} \cdot \frac{1}{J_{w_i}} \right] \\ &= B_s \cdot C_0 \cdot \frac{-\ln(1-Y)}{Y} \cdot \text{CPF} \cdot \sum_{i=1}^N \left[\frac{-\ln(1-Y_i) \cdot \text{CPF}_i}{-\ln(1-Y) \cdot \text{CPF}} \cdot \frac{1}{J_{w_i}} \right] \\ &= B_s \cdot C_0 \cdot \frac{-\ln(1-Y)}{Y} \cdot \text{CPF} \cdot \sum_{i=1}^N \left[w_i \cdot \frac{1}{J_{w_i}} \right] \end{aligned} \quad (\text{A.1})$$

where

$$w_i = \frac{-\ln(1-Y_i)}{-\ln(1-Y)} \cdot \frac{\text{CPF}_i}{\text{CPF}} \quad (\text{A.2})$$

is the weight factor, and its sum

$$\sum_{i=1}^N w_i \approx \sum_{i=1}^N \frac{-\ln(1-Y_i)}{-\ln(1-Y)} = 1 \quad (\text{A.3})$$

is approximately unity. The sum is not exactly 1 due to variations in CPF_i , but the deviation is generally less than 1% when Y_i lies within typical design ranges (e.g., 10%–20% for BWRO at 80% recovery and 6%–15% for SWRO at 50% recovery; see supplementary material for examples). Therefore, the expression within square brackets in Eq. (A.1) may be interpreted as the weighted harmonic mean of J_{w_i} . Moreover, since $-\ln(1-Y_i) \approx Y_i$ for $Y_i < 0.2$, the weight factor can be approximated as:

$$w_i \approx \sum_{i=1}^N \frac{-\ln(1-Y_i)}{-\ln(1-Y)} \approx \frac{Y_i}{\sum_{j=1}^N Y_j} \quad (\text{A.4})$$

whose sum is 1.

The above formulation can, in principle, be extended to N segments in a single element or pressure vessel, where N approaches infinity in the continuous limit. With N goes as high as infinity.

Now consider an RO system divided into N equal-area segments, as shown in Fig. A.11.

It can be shown that:

$$\frac{Y_1}{J_{w_1}} = \frac{Y_2(1-Y_1)}{J_{w_2}} = \frac{Y_3(1-Y_1)(1-Y_2)}{J_{w_3}} = \dots = k \quad (\text{A.5})$$

where k is a constant. Therefore:

$$\begin{aligned} 1-Y_1 &= 1 - kJ_{w_1} \\ (1-Y_1)(1-Y_2) &= 1 - k(J_{w_1} + J_{w_2}) \\ &\vdots \end{aligned} \quad (\text{A.6})$$

$$\prod_{j=1}^i (1-Y_j) = 1 - \sum_{j=1}^i kJ_{w_j}$$

Solving for Y_i :

$$1-Y_i = \frac{1 - \sum_{j=1}^i kJ_{w_j}}{1 - \sum_{j=1}^{i-1} kJ_{w_j}} \quad (\text{A.7})$$

or equivalently:

$$Y_i = \frac{kJ_{w_i}}{1 - \sum_{j=1}^{i-1} kJ_{w_j}} \quad (\text{A.8})$$

When the flux J_{w_i} decreases linearly or quasi-linearly with i , a common trend resulting from increasing osmotic pressure and pressure

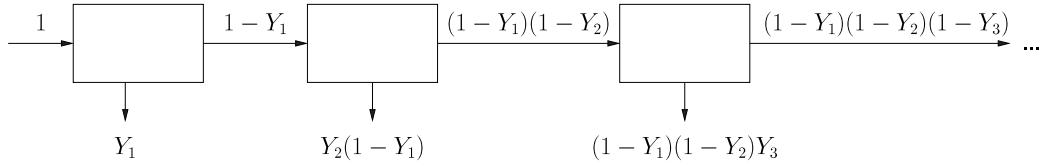


Fig. A.11. Illustration of segment-wise recovery.

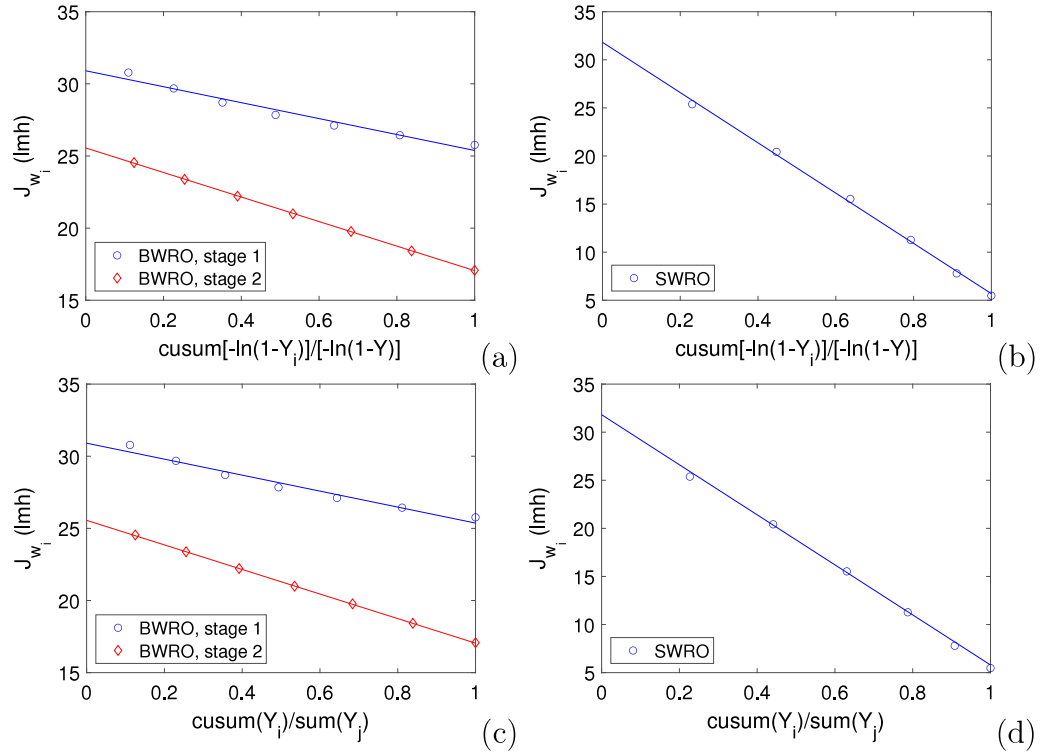


Fig. A.12. (a, c) Linear relationships between the normalized cumulative sums $\sum_{j=1}^i [-\ln(1-Y_j)] / \sum_{j=1}^N [-\ln(1-Y_j)]$ and $\sum_{j=1}^i Y_j / \sum_{j=1}^N Y_j$ versus local flux J_{w_i} for BWRO at 80% recovery; (b, d) corresponding plots for SWRO at 50% recovery. All data are based on ROSA simulation results presented in Section 3.5. “cusum” in the plots denotes cumulative sum.

drop along the membrane length, the cumulative sum $S_i = \sum_{j=1}^i Y_j$ exhibits a linear relationship with J_{w_i} (see Fig. A.12 and more case studies in the supplementary material). For example, consider a mildly curved flux profile expressed as:

$$J_{w_i} = a - b\xi + c\xi^2 \quad (\text{A.9})$$

where $\xi = \frac{i-1}{N-1}$, so that $\xi \in [0, 1]$, and a , b , and c are empirical coefficients with $0 \leq |c| \ll b$ and $0 \leq a$. This profile captures the typical flux behavior in both BWRO and SWRO systems. In this case, the denominator in Eq. (A.8), given by $1 - \sum_{j=1}^{i-1} kJ_{w_j}$, represents the fraction of feedwater not yet recovered. Since J_{w_i} varies smoothly with i , the cumulative sum $\sum J_{w_j}$ also increases smoothly and predictably. As a result, the denominator decreases monotonically and smoothly with i , ensuring that each Y_i remains finite, positive, and continuous under practical operating conditions. Therefore, the cumulative recovery $S_i = \sum_{j=1}^i Y_j$ is a sum of well-behaved terms and is itself a smooth, strictly increasing function. When the flux profile is mildly curved, the total recovery S_i shows quasi-linear dependence on local flux, and can be closely approximated by a linear relation:

$$S_i \approx \alpha J_{w_i} + \beta \quad (\text{A.10})$$

where the constants α and β depend only weakly on the overall system recovery and the curvature of the flux profile.

This quasi-linear behavior is especially accurate when $Y_i < 0.2$, as is typical in seawater and brackish water RO systems. Therefore,

cumulative recovery can be estimated efficiently using local flux measurements, facilitating system analysis and diagnostics.

Now define W_i as the cumulative sum of w_i :

$$W_i = \sum_{j=1}^i w_j = \sum_{j=1}^i \frac{-\ln(1-Y_j)}{-\ln(1-Y)} \quad (\text{A.11})$$

Note that $W_i \propto S_i$, or W_i is quasi-linearly related to J_{w_i} . Thus, in the limit as $N \rightarrow \infty$:

$$\begin{aligned} \lim_{N \rightarrow \infty} \sum_{i=1}^N w_i \cdot \frac{1}{J_{w_i}} &= \lim_{N \rightarrow \infty} \sum_{i=1}^N \frac{W_i - W_{i-1}}{J_{w_i}} \\ &= \int_0^1 \frac{1}{J_w} dW \\ &= \int_{J_{w_1}}^{J_{w_N}} \frac{1}{J_w} \cdot \frac{dW}{dJ_w} dJ_w \\ &= \left[\frac{1}{J_{w_N} - J_{w_1}} \right] \int_{J_{w_1}}^{J_{w_N}} \frac{1}{J_w} dJ_w \\ &= \frac{\ln(J_{w_N}/J_{w_1})}{J_{w_N} - J_{w_1}} \\ &= \frac{1}{J_{w,lm}} \end{aligned} \quad (\text{A.12})$$

This derivation provides a theoretical justification for using the log-mean flux to estimate average permeate concentration in Eq. (26).

Eq. (A.12) shows that the log-mean flux closely approximates the weighted harmonic mean of the local fluxes, provided that the weighting factors sum to unity. The equivalence is exact for linearly decaying flux and remains accurate for mildly curved (quadratic) profiles, as long as J_w varies quasi-linearly with the cumulative weight. This explains the strong agreement between Eq. (26) and detailed simulations across typical BWRO and SWRO configurations.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.memsci.2025.124526>.

Data availability

Data file is provided as supplementary material.

References

- [1] M. Elimelech, W.A. Phillip, The future of seawater desalination: energy, technology, and the environment, *Sci.* 333 (2011) 712–717.
- [2] Y.J. Lim, K. Goh, M. Kurihara, R. Wang, Seawater desalination by reverse osmosis: Current development and future challenges in membrane fabrication—A review, *J. Membr. Sci.* 629 (2021) 119292.
- [3] H. Gu, J. Polanco, K.P. Ishida, M.H. Plumlee, M. Boyd, E. Desormeaux, G.J. Jubry, M.F. Shad, Permeate quality, advanced oxidation process treatability, and cost for two concentrate treatment technologies to enhance recovery for potable reuse, *Water Reuse* 13 (3) (2023) 305–318.
- [4] M.M. Zubair, H. Saleem, S.J. Zaidi, Recent progress in reverse osmosis modeling: An overview, *Desalination* 564 (2023) 116705.
- [5] S. Kumarasamy, S. Narasimhan, S. Narasimhan, Optimal operation of battery-less solar powered reverse osmosis plant for desalination, *Desalination* 375 (2015) 89–99.
- [6] H. Kotb, E. Amer, K. Ibrahim, On the optimization of RO (Reverse Osmosis) system arrangements and their operating conditions, *Energy* 103 (2016) 127–150.
- [7] Z. Hadadian, S. Zahmatkesh, M. Ansari, A. Haghighi, E. Moghimipour, Mathematical and experimental modeling of reverse osmosis (RO) process, *Korean J. Chem. Eng.* 38 (2021) 366–379.
- [8] M. Di Martino, S. Avraamidou, E.N. Pistikopoulos, A neural network based superstructure optimization approach to reverse osmosis desalination plants, *Membr.* 12 (2) (2022) 199.
- [9] I. Alatiqi, A. Ghabris, S. Ebrahim, System identification and control of reverse osmosis desalination, *Desalination* 75 (1989) 119–140.
- [10] L. Gao, A. Rahardianto, H. Gu, P.D. Christofides, Y. Cohen, Energy-optimal control of RO desalination, *Ind. Eng. Chem. Res.* 53 (2014) 7409–7420.
- [11] M. Mulder, *Basic Principles of Membrane Technology*, Kluwer Academic Publishers, Boston, 1997.
- [12] M. Li, J. Li, Incorporating convection into the solution-diffusion framework enhances modeling accuracy in reverse osmosis, *Chem. Eng. Res. Des.* 214 (2025) 441–445.
- [13] Dupont, FilmTec reverse osmosis membranes technical manual, 2020, <https://www.dupont.com/content/dam/dupont/amer/us/en/water-solutions/public/documents/en/45-D01504-en.pdf>. Online; (Accessed 25 April 2020).
- [14] M. Li, Reverse osmosis desalination under periodically oscillating conditions: insight from spatiotemporal simulations, *Desalination* (2025) 118942.
- [15] T.K. Das, M. Folley, P. Lamont-Kane, C. Frost, Performance of a SWRO membrane under variable flow conditions arising from wave powered desalination, *Desalination* 571 (2024) 117069.
- [16] D.R. Paul, Reformulation of the solution-diffusion theory of reverse osmosis, *J. Membr. Sci.* 241 (2) (2004) 371–386.
- [17] P. Biesheuvel, S. Porada, M. Elimelech, J. Dykstra, Tutorial review of reverse osmosis and electrodialysis, *J. Membr. Sci.* 647 (2022) 120221.
- [18] L. Wang, T. Cao, J.E. Dykstra, S. Porada, P. Biesheuvel, M. Elimelech, Salt and water transport in reverse osmosis membranes: Beyond the solution-diffusion model, *Environ. Sci. Technol.* 55 (24) (2021) 16665–16675.
- [19] S. Avlonitis, W.T. Hanbury, M.B. Boudinar, Spiral wound modules performance an analytical solution: Part II, *Desalination* 89 (1993) 227–246.
- [20] M. Li, Predictive modeling of a commercial spiral wound seawater reverse osmosis module, *Chem. Eng. Res. Des.* 148 (2019) 440–450.
- [21] G.E. Dévora-Isiordia, C.A. Cásares-De la Torre, D.P. Morales-Mendivil, R. Montoya-Pizeno, N. Velázquez-Limón, J.A. Aguilar-Jiménez, J. Ríos-Arriola, Evaluation of concentration polarization due to the effect of feed water temperature change on reverse osmosis membranes, *Membr.* 13 (1) (2022) 3.
- [22] Y. Taniguchi, An analysis of reverse osmosis characteristics of ROGA spiral-wound modules, *Desalination* 25 (1978) 71–88.
- [23] M. Li, B. Noh, Validation of model-based optimization of brackish water reverse osmosis (BWRO) plant operation, *Desalination* 304 (2012) 20–24.
- [24] H.-J. Oh, T.-M. Hwang, S. Lee, A simplified simulation model of RO systems for seawater desalination, *Desalination* 238 (1–3) (2009) 128–139.